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AI/ML intern

DAY-6

TASK : 5 Integrating AI Outputs with Backend APIs

The objective of this task was to start integrating AI-generated outputs with backend APIs so that pose detection results, body measurements, and recommendation data could be transferred between the AI model and the backend system in real time.

What I Worked On

During this task, I worked on connecting the AI processing module with backend APIs using Python and Flask. I integrated the pose detection and body measurement outputs generated using MediaPipe and OpenCV with a backend API created using Flask.

The AI model generated outputs such as:

- Shoulder width
- Hip width
- Estimated height
- Pose landmark information

These outputs were converted into JSON format and sent to the backend server using HTTP POST requests.

Integration Process I Followed

1. AI Processing

The webcam feed was processed using MediaPipe Pose Detection to extract body landmarks and estimate measurements.

2. Data Preparation

The generated measurements were stored inside a Python dictionary.

Example:

```
data = {
    "shoulder_width": shoulder_width_cm,
    "hip_width": hip_width_cm,
    "height": body_height_cm
}
```

3. Backend API Creation

I created a backend API using Flask to receive AI-generated data.

4. API Communication

The AI outputs were sent to the backend using HTTP POST requests.

Example:

```
response = requests.post(
    "http://127.0.0.1:5000/bodydata",
    json=data
)
```

5. Backend Response

The backend server successfully received and displayed the JSON data in the terminal.

Technologies Used

- Python
- Flask
- MediaPipe
- OpenCV
- JSON
- HTTP Requests

Observations:

AI outputs were successfully transferred to the backend server.

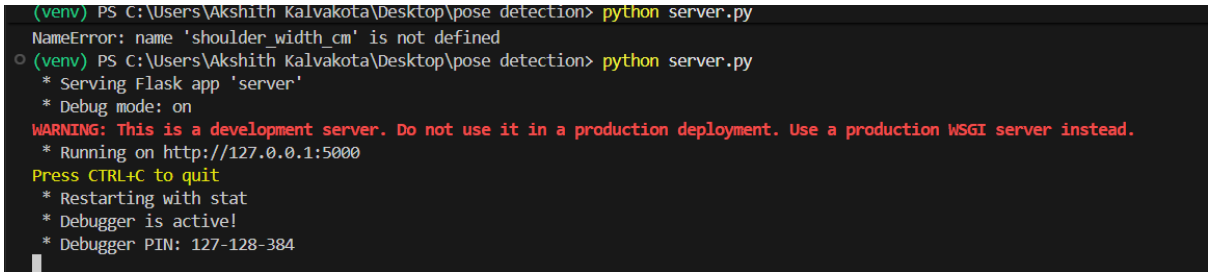
JSON format made data communication simple and organized.

Flask APIs were easy to integrate with Python AI modules.

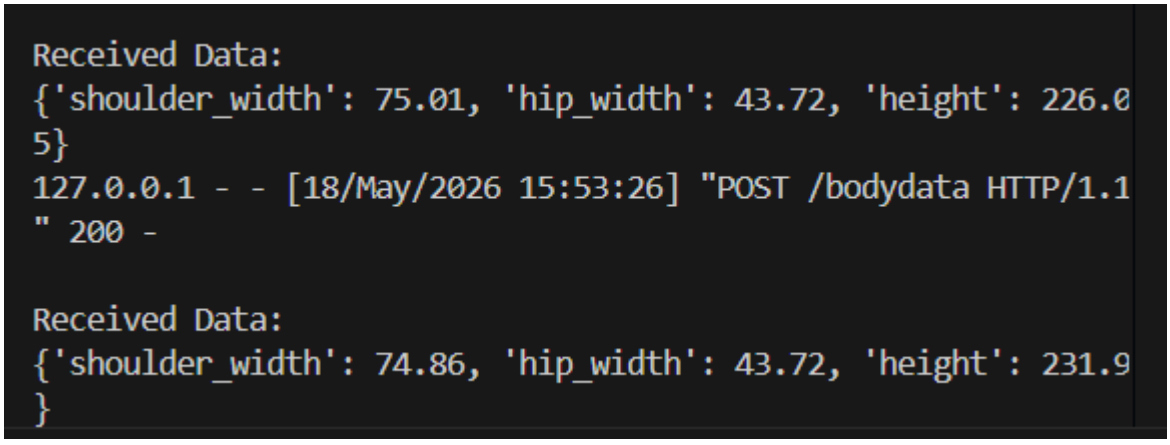
Real-time webcam processing could be connected directly with backend systems.

SCREENSHOTS:

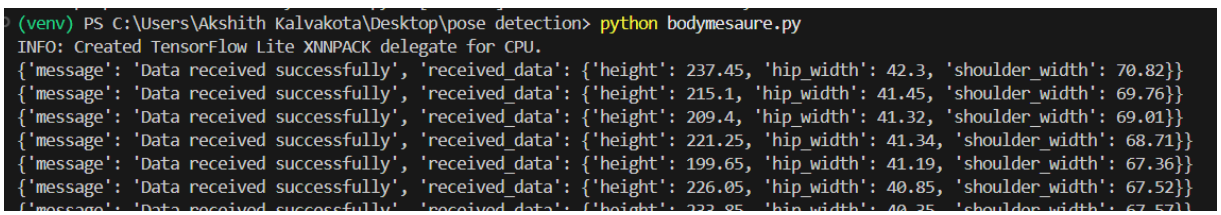
1. **FLASK SERVER RUNNING:**



2. **RECEIVED JSON DATA IS FLASK TERMINAL:**



3. **API SUCCESS RESPONSE:**



What I Learned

Through this task, I learned how AI models communicate with backend systems using APIs. I understood that backend integration is essential for transferring AI-generated outputs into real-world applications such as virtual try-on systems and recommendation platforms. I also learned how JSON data and HTTP requests are used to exchange information between AI modules and backend servers.